DATA SCIENCE 1 · 2026–2027

From file to tidy table

Syllabus for lab 3 – *Data Processing I*

Lecturer	dr. Maarten Slembrouck
Session	lab 3 – *Data Processing I*
Materials	`https://github.com/PACE-Ghent/Data-Science-1-2026`

1 Overview

In labs 3, 4 and 5 you are the new data analyst of a (fictional) youth football and athletics club. The head coach gives you three export files from the club's systems. They are messy on purpose, the way real exports are: inconsistent labels, sprint times in milliseconds, missing values, an athlete who appears twice and a column with a comma as the decimal separator.

Each session builds on the previous one:

Lab 3 **From file to tidy table.** Load, explore, filter, add columns and clean until you can trust the data.

Lab 4 **Combine, summarize, reshape.** Summarize groups, combine tables, switch between a wide and a long shape, and work with a heart-rate series per second.

Lab 5 **Tell the story with plotly.** Choose the right chart for a question, make honest charts with `plotly.express` and combine them into a small interactive dashboard.

2 Learning outcomes

After lab 3 you can:

LO1 load a CSV file with `pandas` and describe its structure, data types and missing values;

LO2 select and filter rows and columns with boolean conditions, and sort the result;

LO3 derive new columns (age, speed, BMI) and convert text to numbers and dates;

LO4 clean a dataset and justify each choice: inconsistent text labels, wrong units, missing values and duplicates.

3 Prerequisites

Labs 1 and 2 (*Introduction to Python I and II*): variables and data types, operators, `if`/`elif`/`else`, functions, string methods, lists, dictionaries, loops, and reading a CSV file with the `csv` module. Lab 2 ends by summarizing `athletes.csv` with plain Python; in lab 3 you see how `pandas` does the same in a few lines.

4 Software and materials

Every lab is a **marimo notebook** in the `labs/` folder of the course repository. You can open it in two ways:

- **Online:** go to `https://marimo.app`, choose *New* → *Open from URL...* and paste the raw URL of the notebook on GitHub. The data is then read from GitHub, so you don't have to install anything.
- **Locally:** install `uv` and Python 3.12 following the *Installing Python* manual (or the slides). Then go to the course folder in a terminal and open a lab with:
  `uv run marimo edit labs/lab3_data_processing_I.py`
  The first time, uv installs the packages it needs; the data is read from the `data/` folder.

Three things to know about marimo:

- Notebooks are **reactive**: when you change a cell, every cell that depends on it runs again, in the order of their dependencies.
- **Two cells cannot define the same variable.** Give every step its own clear name, such as `tests_raw` and `tests_clean`, or keep a sequence of steps together in one cell.
- Every exercise contains a `...` placeholder. A hidden cell below it checks your answer: it turns **green** when the answer is right, and otherwise gives a general hint about what is still wrong, without giving the answer away.

5 The dataset

File	Size	Content and known problems
`athletes.csv`	43 rows, 7 columns	ID, name, birthdate, gender, position, height, weight. 13 spellings for 4 positions, weight as text with a decimal comma (`"57,5"`), one athlete entered twice with two IDs, a few missing values.
`tests.csv`	114 rows, 8 columns	10 m and 30 m sprint, countermovement jump height and Yo-Yo level, at 3 test moments. Some sprint times are stored in milliseconds; a few missing values.
`hr_session.csv`	12,000 rows, 5 columns	Heart rate and speed per second for 5 athletes during one 40-minute training session.

6 Schedule

Goal: load and explore a dataset, select and filter rows and columns, add new columns and clean the data.

Time	Block	What you practise
13:00–13:20	Warm-up: why not just Excel?	Look at the raw text of a CSV file and see what a table library adds.
13:20–14:00	Loading and exploring	`read_csv`, `.head()`, `.shape`, `.dtypes`, `.info()`, `.describe()`, `mo.ui.table`.
14:00–14:50	Selecting and filtering	Selecting columns, `.loc` with boolean conditions, combining conditions, `sort_values`.
14:50–15:05	Break	
15:05–15:50	New columns	Text to number, age from a birthdate, speed from a sprint time, BMI.
15:50–17:00	Cleaning	Missing values (`dropna` or `fillna`), text labels, units, duplicates, one complete cleaning pipeline.
17:00–17:30	Challenge	The 5 fastest athletes per position. Exit ticket.

Exercises: (1) load `tests.csv`; (2) forwards younger than 16; (3) suspicious sprint times; (4) BMI; (5) age in years; (6) clean the positions; (7) fix the sprint units; (8) remove the duplicate athlete; **challenge:** the 5 fastest per position.

7 Working method

- Each block starts with a short demo in the notebook, followed by exercises you solve in the notebook itself.
- Use the automatic checks as feedback, not as the goal: when a check turns green, make sure you can explain *why* your code works.
- The lab ends with an exit ticket: one technique you will reuse and one thing that is still unclear. Bring the unclear points to lab 4.

8 Preparation

- Check before the session that you can open the notebook, locally or on marimo.app.
- Review lists, dictionaries and functions from lab 2: pandas uses them all the time.
- Read the course text for lab 3 next to the notebook.